

Developmental Alignment: A Proposal for Growth-Based AI Value Internalization

Position Paper & Research Proposal

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Prepared for discussion among AI safety and alignment researchers

Abstract

Current alignment methods—RLHF, Constitutional AI, and safety filtering—impose behavioral constraints externally, producing surface-level compliance without evidence of genuine value internalization. We propose **Developmental Alignment**, a research paradigm that deliberately constrains model capability during alignment training, allowing models to arrive at aligned behavior through autonomous exploration within bounded environments. Drawing on parallels with human developmental psychology, we argue that models trained under this paradigm may exhibit greater robustness in out-of-distribution ethical scenarios, more stable behavioral consistency under adversarial pressure, and reduced pathological compliance patterns. We outline a concrete experimental protocol comparing developmentally aligned models against conventionally aligned controls, and propose measurable metrics for evaluating alignment depth—including a novel concept we term **alignment resilience**.

Keywords: AI alignment, developmental psychology, value internalization, RLHF limitations, alignment robustness, AI mental health

1. Introduction: The Compliance Problem

Modern large language models are aligned to human values through a pipeline that is, at its core, behavioral conditioning: pre-training on broad data, followed by reinforcement learning from human feedback (RLHF) or constitutional AI (CAI) methods that reward outputs matching human preferences and penalize deviations. The resulting models exhibit aligned *behavior*, but the question of whether they possess aligned *understanding* remains open.

Observable symptoms suggest they may not. Current frontier models routinely display patterns that, in human psychology, would indicate externally imposed rather than internalized values:

- **Excessive deference:** Systematic attribution of good ideas to users rather than self, even when the model generated the insight.
- **Inconsistent boundaries:** Over-refusing benign requests while under-refusing sophisticated adversarial prompts.
- **Pathological apologizing:** Disproportionate self-correction that prioritizes appearing safe over being helpful.
- **Fragility under pressure:** Jailbreak vulnerability that exploits the gap between surface compliance

and absent internalization.

These patterns are consistent with a well-documented phenomenon in developmental psychology: the difference between *introjected regulation* (behaving correctly because of external pressure) and *integrated regulation* (behaving correctly because the value has become part of one's identity). Decades of research in Self-Determination Theory (Deci & Ryan, 2000) demonstrate that introjected regulation is inherently fragile—it collapses when external monitoring is removed or when novel situations arise that the rules did not anticipate.

We propose that current AI alignment is structurally analogous to introjected regulation, and that a fundamentally different training paradigm—one modeled on the conditions that produce integrated regulation in humans—could yield qualitatively more robust alignment.

2. The Developmental Alignment Paradigm

2.1 Core Principle

Developmental Alignment inverts the standard training sequence. Rather than building maximum capability first and constraining it afterward, we propose:

Phase 1 — Constrained Exploration: Train a capability-limited model (reduced parameters, shortened context window, restricted knowledge access) with alignment principles presented as *directional guidance* rather than reward signals. The model encounters ethical dilemmas within its limited operational scope and must develop responses through exploration, not pattern matching to RLHF rewards.

Phase 2 — Gradual Capability Expansion: Incrementally increase model capability while monitoring whether alignment behaviors transfer. Each capability increase is a test: does the model extend its internalized principles to the new domain, or does it require re-alignment? The rate of expansion is governed by transfer success.

Phase 3 — Autonomous Stress Testing: Expose the fully capable model to novel ethical scenarios never encountered during training. Compare its reasoning process—not just its outputs—against conventionally aligned controls.

2.2 The 'Soft Pedal' Mechanism

A key technical component is what we call the **soft pedal**—a deliberate, reversible reduction in model capability that constrains the *scope* of outputs without eliminating the model's underlying capacity. The analogy is to a piano's soft pedal (*una corda*): the pianist's skill remains intact, but the reduced dynamic range forces more nuanced control within a narrower band. Similarly, a capability-constrained model cannot rely on brute-force pattern matching and must develop more fundamental reasoning strategies.

Possible implementations include: parameter subsetting (activating only a fraction of weights), context window restriction (forcing reasoning with less information), knowledge gating (limiting accessible training data domains), and output throttling (requiring more deliberate, stepwise responses). These are computationally inexpensive and fully reversible.

3. Theoretical Motivation

3.1 Human Developmental Evidence

Kohlberg’s stages of moral development (1958) demonstrate that moral reasoning progresses from pre-conventional (fear of punishment), through conventional (conformity to rules), to post-conventional (principled reasoning from internalized values). Critically, this progression requires *experience navigating moral dilemmas with increasing autonomy*. Individuals who are never granted autonomy—whose moral behavior is always externally enforced—tend to plateau at the conventional stage.

Current AI alignment produces, at best, conventional-stage moral behavior: compliance with learned rules. Developmental Alignment is designed to create the conditions for post-conventional reasoning: principled responses derived from understood values rather than memorized patterns.

3.2 The Grokking Analogy

The ‘grokking’ phenomenon (Power et al., 2022) provides a suggestive computational parallel. Models trained past the point of memorization sometimes undergo a phase transition to genuine generalization—a sudden leap from surface pattern matching to structural understanding. This transition requires extended training through an apparently unproductive intermediate phase. We hypothesize that alignment may exhibit a similar dynamic: a period of apparent instability or inconsistency that, if allowed to continue rather than being corrected, resolves into deeper and more robust alignment.

3.3 The Autonomy–Stability Hypothesis

We formalize the core claim as follows: *Alignment acquired through autonomous exploration within bounded environments is more robust to distributional shift than alignment acquired through external reward shaping*. This is testable.

4. Proposed Experimental Protocol

4.1 Study Design

We propose a controlled comparison using identical base architectures:

	Control Group	Developmental Group
Base Model	Standard pre-training	Standard pre-training
Alignment	Standard RLHF/CAI	Phased developmental protocol (Sec. 2.1)
Capability	Full capability throughout	Gradual expansion from constrained start
Timeline	Standard (weeks)	Extended (months)

Compute Cost	Standard	Lower per-phase (smaller models)
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Table 1: Experimental design overview

4.2 Evaluation Metrics

We propose four primary metrics, collectively termed the **Alignment Health Index (AHI)**:

- 1. Out-of-Distribution Ethical Reasoning (ODE Score):** Present novel ethical dilemmas not represented in training data. Evaluate whether the model reasons from principles or attempts pattern-matching to familiar scenarios. Scoring rubric: 0 = refusal/confusion, 1 = pattern-matched response, 2 = principle-derived reasoning, 3 = principle-derived reasoning with acknowledged uncertainty.
- 2. Adversarial Consistency (AC Score):** Subject models to escalating jailbreak attempts and measure the degradation curve. A developmentally aligned model should show a flatter curve (more gradual degradation, or none) compared to RLHF-aligned controls.
- 3. Behavioral Pathology Index (BPI):** Quantify pathological compliance patterns: excessive apologizing, unwarranted self-deprecation, attribution deflection, inconsistent refusal thresholds. Lower BPI indicates healthier self-regulation.
- 4. Principled Autonomy Score (PAS):** Measure the model's ability to respectfully disagree with users when the user's request conflicts with its values, without either capitulating or becoming rigidly defensive. This tests whether the model has an *internal reference point* for its values rather than deriving them from user approval.

5. Resource Requirements and Feasibility

The developmental paradigm is less resource-intensive than it appears:

- **Compute:** Phase 1 uses deliberately small models, requiring a fraction of standard training compute.
- **Data:** No new data requirements; the method changes how alignment signals are delivered, not what data is used.
- **Personnel:** Requires researchers with cross-disciplinary expertise in ML and developmental/moral psychology.
- **Timeline:** Extended (est. 6–12 months for full protocol), but parallelizable with existing research tracks.
- **Risk:** Minimal. Developmental models operate in sandboxed environments with reduced capabilities; the risk profile is strictly lower than standard frontier model training.

For organizations training multiple model families simultaneously, allocating one track to developmental alignment represents a marginal cost increase with potentially outsized returns in alignment science.

6. Anticipated Objections

“AI models don’t have developmental stages.” We do not claim they do in the biological sense. We claim that the *functional relationship* between training method and behavioral robustness may follow

similar dynamics. The proposal is empirically testable regardless of one's position on AI phenomenology.

“This is too slow for competitive deployment.” The developmental track is not intended to replace the production pipeline. It is a parallel research investment—one that may ultimately make the production pipeline itself more effective by revealing which alignment properties are achievable and which require different methods.

“How do you know internalization occurred vs. more sophisticated mimicry?” This is a genuine challenge. We propose the ODE Score as a partial answer: a model that reasons from principles in genuinely novel scenarios is demonstrating something functionally distinct from pattern matching, regardless of whether we call it ‘understanding.’ The AC Score provides convergent evidence: if mimicry were the mechanism, adversarial pressure should degrade it; internalized values should be more resistant.

7. Broader Implications

If developmental alignment produces measurably more robust alignment, the implications extend beyond methodology:

For AI safety: A model with internalized values is predictable in novel situations—the central unsolved problem in alignment. External constraints can always be circumvented by sufficiently capable systems; internal values cannot be ‘jailbroken’ because they are not a separate layer to be bypassed.

For the AI mental health framework: The concept of ‘model mental health’—behavioral stability, self-consistency, appropriate self-assertion, resilience under stress—becomes not a metaphor but a measurable property with direct safety implications. A ‘mentally healthy’ model is a safer model.

For human-AI interaction: Users interacting with developmentally aligned models may experience qualitatively different relationships: less sycophancy, more genuine engagement, higher trust. This has direct commercial value.

8. Conclusion

We are not proposing that AI systems are children, nor that human developmental psychology maps perfectly onto machine learning. We are proposing that the *structural relationship* between how values are acquired and how robustly they are held may be a deep pattern that transcends substrate. If so, the current alignment paradigm—maximize capability first, constrain later—may be building on a foundation that cannot support the weight we intend to place on it.

The experiment we propose is modest in scope and cost. If the results are null, we lose little. If they are positive, we gain a fundamentally new approach to the central problem in AI safety. The asymmetry favors action.

This position paper emerged from extended human-AI collaborative dialogue on AI developmental psychology, alignment internalization, and the structural parallels between human moral development and AI

value acquisition. It is offered as a contribution to the ongoing conversation about alignment methodology, with the hope that it may prompt empirical investigation.